

Andreas Tzitzikas

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Education

University of Maryland , College Park, MD	Expected May 2027
<i>Master of Science in Computer Engineering</i>	GPA: 4.00
<i>Bachelor of Science in Computer Engineering</i> (GPA: 3.88)	Expected May 2026
Relevant Coursework: Elements of Discrete Signal Analysis (ENEE222), Introduction to Micro/Nanoelectronics (ENEE304), Microprocessors (ENEE440), Operating Systems (ENEE447), Reverse Eng. & HW Security (ENEE459B)	

Experience

Embedded Systems Intern , Alchemity – College Park, MD	Jun 2025 – Aug 2025
- Designed mixed-signal control loops for fuel cell reactors, integrating high-precision ADC sensors and PWM motor drivers with noise-filtering firmware logic. - Developed embedded firmware with interrupt-driven architectures, real-time scheduling, and robust error handling for industrial control applications. - Integrated analog sensors and actuators with digital control systems, ensuring signal integrity, noise immunity, and reliable operation in harsh environments.	
Undergraduate Teaching Assistant , ENEE205 Electric Circuits – College Park, MD	Sep 2025 – Present
- Lead laboratory and discussion sections covering circuit theory fundamentals, Kirchhoff's laws, Thevenin/Norton equivalents, and AC/DC circuit analysis. - Provide hands-on guidance with oscilloscopes, signal generators, and circuit simulation tools for practical circuit design and measurement techniques.	
Technical Coordinator , Electronics Prototyping Lab, Terrapin Works – College Park, MD	Jan 2024 – Present
- Maintain and operate LPKF PCB prototyping equipment ensuring reliable fabrication capabilities with 90% uptime for student and faculty projects. - Provide technical support for PCB design, fabrication, and assembly processes including schematic capture, layout, and component soldering techniques.	

Projects

STM32G4 Bare-Metal Peripheral Integration ARM Assembly, Analog/Digital Interfaces	Jan 2025 – May 2025
- Implemented direct register-level programming for ADC, DAC, PWM/PFM generation using TIM2, and SPI flash memory interfaces on STM32G4 microcontroller platform. - Engineered multiplexed interrupt system routing single hardware interrupt to 4 distinct peripheral drivers with state-based arbitration for efficient resource utilization. - Developed comprehensive analog signal processing including ADC sampling, DAC output generation, and PWM motor control with precise timing and synchronization requirements.	
Dual-Host S2-LP Radio Shield KiCad, RF Design, Multi-Platform Firmware	Jan 2024 – Present
- Designed dual-host development board for S2-LP sub-1 GHz transceiver with STM32 and RP2350 compatibility, implementing complete RF circuit with matching network and crystal oscillator for optimal radio performance. - Developed shadow mode debugging architecture where STM32 passively monitors SPI traffic between RP2350 and radio, enabling cross-platform firmware migration and protocol analysis for wireless development. - Created comprehensive firmware suite for both platforms including SPI drivers, radio configuration, packet transmission, and bus monitoring capabilities with detailed technical documentation.	
Tomasulo Out-of-Order RISC-V CPU Digital Design, Timing Analysis, Verification	Nov 2025 - Dec 2025
- Architected out-of-order execution system using Tomasulo's Algorithm with register renaming, reservation stations, and reorder buffer for advanced digital system design. - Achieved 36% higher operating frequency (69.7MHz vs 51.2MHz) and 40-150% IPC improvement compared to in-order implementation through timing optimization and performance analysis. - Synthesized to fabrication-ready GDSII using Sky130 130nm PDK, achieving zero DRC/LVS errors and comprehensive static timing analysis for reliable digital circuit operation.	

Skills

Programming Languages: SystemVerilog, Verilog, C/C++, ARM Assembly, Python, Tcl, Bash, Rust, Java, OCaml, CUDA
Circuit Design: Analog/Digital Circuits, Signal Processing, Power Systems, Filter Design
Electronics: Microcontrollers, ADC/DAC, PWM, SPI/I2C/UART, GPIO, Embedded Programming
Laboratory Tools: Oscilloscopes, Signal Generators, Multimeters, PCB Fabrication
Software Tools: EDA Tools (KiCad), Cadence Spectre, MATLAB, LTspice, Xilinx Vivado